Lightweight Cross-Modal Information Measure and Propagation for Road Extraction From Remote Sensing Image and Trajectory/LiDAR

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***Abstract*— Recent studies have confirmed that GPS trajectory can effectively assist in achieving more accurate road extraction from remote sensing images (RSIs). Therefore, lots of efforts focus on designing effective multimodal fusion strategies for GPS trajectory and RSI modalities. However, there are still some limitations, e.g., the fusion structures are complex and hinder further improvements. Moreover, the negative impact of redundant information in various modalities is commonly ignored. This article aims to design a simple yet effective fusion strategy for GPS trajectory and RSI modalities to address the above issues. Inspired by the network pruning algorithm, we design a cross-modal information propagation (CMIP) mechanism. CMIP utilizes the scaling factors and sparse constraint to distinguish the redundant information of a certain modality that is directly replaced with corresponding information of another modality. We improve the widely used $\mathcal{L}1$ sparse constraint and propose a novel information balanced constraint which is added on the scaling factors to better identify and prune redundant channels. Embedded in the CMIP mechanism, a multimodal information propagation network, CMIP network (CMIPNet) is proposed, which can fully explore the complementarities between different modalities to accurately locate roads, especially roads with noise or incomplete information of a certain modality. Since the CMIP is parameter-free and self-adaptive, CMIPNet is lightweight and easy to deploy. The parameter number of CMIPNet can be comparable to single-modal models, which is about one-third of the existing multimodal models. Extensive experiments are performed to demonstrate that CMIPNet outperforms the previous single- and multimodal road extraction methods.**

***Index Terms*— Channel propagation, multimodal road extraction, remote sensing image (RSI), sparse constraint.**

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Data is available on-line at https://github.com/yuqiuyuqiu/CMIPNet.

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I. Introduction

AUTOMATICALLY extracting traffic roads from remote sensing images (RSIs) has extremely important economic value and scientific research significance in map update, urban road planning, vehicle intelligent navigation, and so on [1] and [2]. Early models [3], [4], [5] heavily rely on handcrafted features and heuristic priors like color contrast, texture, and contour prior, coupled with the shallow models such as support vector machines [6] and Markov random fields [7] for road region recognition. However, the handcraft features can hardly capture high-level semantic information which is important to locate roads especially in complicated traffic scenes, limiting the applicability of these approaches.

In recent decades, with the fast development of deep learning, especially convolutional neural networks (CNNs), numerous CNNs-based algorithms have been proposed to achieve computer-aided traffic road extraction. Benefiting from the powerful representation learning capabilities of CNNs, these road extraction models continuously push forward the state of the art and bring great progress [8], [9], [10], [11], [12]. For example, Zhang et al. [13] combined U-Net and residual learning architecture to recognize road regions. Zhou et al. [14] expanded the receptive field of LinkNet [15] with dilated convolutions, and proposed the D-LinkNet model for road extraction from high-resolution RSIs. However, RSIs significantly differ from natural images, whose characteristics include diverse spatial resolutions, complex background structure and topology, distinct imaging orientations, the fuzziness and thinness of the roads, and more complex visual attributes of the roads covered by trees [16]. It seriously hinders the performance of extracting roads from RSIs. What's worse, some infrastructure elements (e.g., loess and train tracks) resemble the roads, introducing extraction complexity without contextual cues. Hence, it is still a challenging problem to accurately segment traffic roads.

To address the above issues, many studies have introduced other modality data, such as GPS vehicle trajectory and light detection and ranging (LiDAR) data, to assist in achieving more accurate road extraction from RSIs [17], [18], [19], [20], [21], [22], [23]. For the trajectory data, regions with numerous vehicle trajectory signals tend to correlate with road areas. For example, Sun et al. [17] collected two taxi and bus GPS datasets from Beijing and Shanghai, and integrated trajectory

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heatmaps and RSIs into the network to enhance road prediction. Liu et al. [18] introduced a dual-enhancement module that fostered interaction between trajectory and images. Besides, LiDAR data offers depth and distance information, i.e., different objects reflect different laser signals. It can highlight road flatness effectively, aiding in distinguishing road areas from buildings and trees. Consequently, lots of techniques [20], [21], [22], [23] have been proposed for recognizing road regions by jointly utilizing image and LiDAR data. Although significant progress has been achieved in this direction, the fusion structures have become more and more complex and heavyweight, which will make the future improvements difficult and heavily engineered. Moreover, the manually designed fusion strategies would be suboptimal due to the large search space of possible solutions.

In this article, we attempt to build a simple yet effective multimodal fusion strategy, which makes full use of the valuable information and reduces the negative impact of noise in different modalities. In detail, we propose a cross-modal information propagation (CMIP) mechanism which utilizes the scaling factors and sparse constraint to force the network to distinguish the redundant information of a certain modality and directly replace it with corresponding information of another modality. The main architecture of the CMIP mechanism is shown in Fig. 1. The core concept of CMIP is based on the assumption that smaller-norm channels are less informative, drawing inspiration from network pruning algorithms [24], [25]. To clarify, we employ the scaling factor (γ in this article) of the batch-normalization (BN) layer [26] as a metric to measure the importance of each corresponding channel. Channels with scaling factors close to zero of a modality are then replaced with the corresponding channels of another modality. Moreover, we improve the $\mathcal{L}1$ sparse constraints on scaling factors to better identify and prune redundant channels. We verify that the commonly used $\mathcal{L}1$ constraint tends to push all scaling factors toward zero including those with relatively large values, because the optimal solution of $\min \|\gamma\|_1$ is $\mathbf{0}$. Hence, to address the above limitation of $\mathcal{L}1$ constraint, we design an information balanced constraint, which can drive the smaller scaling factors toward zero while exerting minimal constraints on the larger values. Note that this information propagation process operates without the need for additional parameters, and it adapts itself dynamically through the control of scaling factors during the training process. By using the CMIP module to achieve the exchange of useful information from various modality encoders, we propose a multimodal information propagation network, namely CMIPNet. Overall, CMIPNet can fully explore the complementarities between different modalities to accurately locate the road, especially when dealing with the noise or incomplete road information of a certain modality. We summarize our contributions as follows.

1) We design a self-adaptive CMIP mechanism which can achieve simple, efficient, and effective interaction of image modality and trajectory/LiDAR data, and dynamically reduce the negative influence of intra-modal noises.
2) For CMIP mechanism, we improve the $\mathcal{L}1$ sparse constraint and propose an information balanced constraints on scaling factors to better identify and prune redundant channels.
3) Embedding CMIP, we propose a unified multimodal road extraction framework for accurate road extraction. It is lightweight making it easy to deploy and develop.
4) We conduct extensive experiments to demonstrate that 1) the parameter number of our CMIPNet can be comparable to that of the single-modal models, which is approximately one-third of the existing multimodal models and 2) the performance of CMIPNet outperforms the previous single- and multimodal road extraction methods.

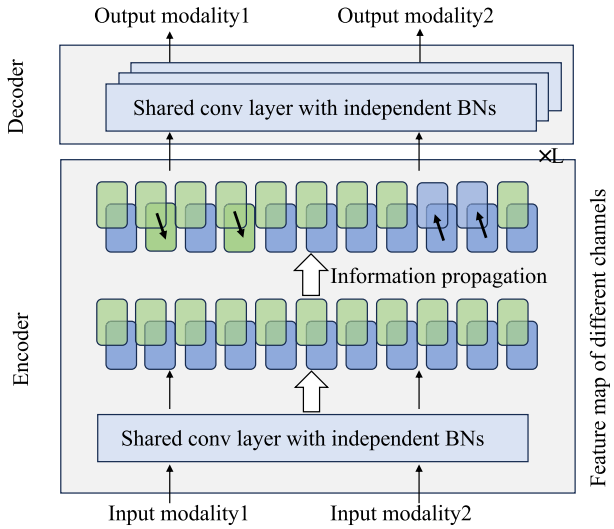


Fig. 1. Main idea of this work, i.e., the information propagation process guided by the independent scaling factors in BN layers.

The remainder of our article is organized as follows. Section II introduces the related works on road extraction from various modalities. Section III describes our CMIPNet in detail. Section IV describes the implementation details and experimental results. Section V draws the conclusions of this work.

II. Related Work

The previous works on automatic road extraction can be divided into two categories: single- and multimodal-based methods. In this section, we provide a brief overview of these two methods.

A. Single-Modal-Based Method

The previous approaches can be classified into three categories based on the type of input data.

1) Road Extraction From RSIs: In the era of industrialization, the rapid evolution of remote sensing technology has made it easy to obtain a large number of high-quality RSIs. Early works [27], [28], [29], [30] heavily rely on manual feature engineering such as textures and outlines, which are then fed into the shallow models (e.g., Markov random fields) to extract roads. However, the manual features can hardly capture high-level semantic information which limits the ability of models to accurately extract roads, especially in

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complicated traffic scenes. More recently, the development of deep neural networks, especially CNNs [31], has driven this field benefiting from their powerful multiscale representation learning capabilities. For instance, Cheng et al. [9] introduced a cascading CNN to simultaneously segment roads and extract road centerlines. Zhang et al. [13] combined residual learning and U-Net architecture for semantic segmentation to recognize the road regions. Zhou et al. [14] expanded the receptive field of LinkNet [15] with dilated convolutions, and proposed the D-LinkNet model for road extraction from high-resolution RSIs. Fu et al. [32] used multiscale fully convolutional networks (FCNs) to predict pixel categories, and then refined outputs through conditional random field post-processing. Yang et al. [33] leveraged road position information to enhance foreground contextual information, and combined the contextual information with the same resolution feature map from encoders to supplement accurate road structure features for decoder stages. Despite progress, many challenges still remain, particularly in complex and extreme occlusion scenarios. In conclusion, exclusively relying on visual data from RSIs to extract roads is challenging. Therefore, complementary information from other sources could be introduced for more effective extraction.

2) Road Extraction From GPS Trajectories: Intuitively, regions with numerous vehicle trajectory signals tend to correlate with road areas. This principle inspires lots of researchers to explore road extraction from crowdsourced trajectory data. Because the GPS data has excessive noise in both variation and bias, and its high differences in disparity in density, resolution, and distribution, previous studies mainly put their effort into mitigating GPS noise and uncertainty. Conventional approaches fall into four categories. First, clustering-based models [34], [35], [36] formulate road identification as a network alignment optimization challenge, where both road nodes and edges are inferred through spatial clustering. Second, trajectory merging models [37], [38] either incorporate GPS trajectories into the existing road segments or generate new segments when matches are lacking. Third, kernel density estimator-based methods [39], [40] transform GPS trajectories into density maps via kernel density estimation [41], and then employ image processing for centerline extraction. Finally, deep neural network-based models have been introduced into this field recently. For example, Ruan et al. [42] proposed a deep learning-based framework for inferring road centerlines from trajectory data in the spatial and transitional views. Despite lots of techniques, limitations still exist due to the constrained crowdsourced trajectory information and challenges in GPS noise reduction, resulting in far from satisfactory road extraction results from GPS Trajectories.

3) Road Extraction From LiDAR: Compared to aerial imagery, LiDAR data provides depth and distance information, and different laser signals reflected by different objects (such as buildings, trees, and roads). This characteristic of LiDAR data allows road flatness to stand out from an aerial perspective, aiding the differentiation of road proposals from buildings and trees. Lots of scholarly studies have produced algorithms [43], [44], [45] for identifying roads using LiDAR data. For example, Hu et al. [46] first filtered non-ground LiDAR points, then used multiple features to detect road centerlines from the remaining ground points. Despite progress, road extraction based on LiDAR data continues to pose challenges. Current approaches encounter difficulties in complex scenarios due to sparse LiDAR data and noise artifacts [47].

B. Multimodal-Based Method

As above mentioned, each modality of data contains useful and useless (even noisy) information. Therefore, combining their complementary useful information is a naive approach for efficient road extraction. In this direction, numerous techniques have been suggested for recognizing road regions by utilizing both RSIs and LiDAR data. For instance, in the work of Hu et al. [22], road primitives were first segmented from RSIs and LiDAR data. Subsequently, road stripes were detected using an iterative Hough transform algorithm. Finally, the final road network was formed through topology analysis. Parajuli et al. [21] employed two SegNet [20] models to independently extract features from RSIs and LiDAR data, then used another SegNet combined modular features to predict ultimate road maps.

Nonetheless, either remote sensing imagery or LiDAR data in isolation falls short of providing adequate information for the identification of traffic roads that are significantly obstructed by trees. Due to the rapid advancement of mobile platforms and localization techniques, it is now much easier to collect these GPS trajectories. Consequently, we can readily utilize them to build a substantial trajectory database [48], [49], [50] and convert it into input data that the model can accept. Hence, recent approaches have integrated RSIs and vehicle trajectories. For instance, Sun et al. [17] integrated trajectory heatmaps and RSIs into various backbone networks, such as U-Net [51], Res-UNet [52], LinkNet [15], and D-LinkNet [14], to facilitate traffic road prediction. For example, Zhang et al. [53] trained the CNN models by the dataset with GPS trajectories for extracting roads from satellite imagery to make full use of the complementary characteristics of floating car data and high-resolution. Liu et al. [54] proposed a multitask and multisource adaptive fusion network, which leveraged RSIs and trajectory datasets to execute road extraction and intersection detection tasks simultaneously. Specially, an adaptive fusion module was created to fuse the individual road features at each level in a guiding fashion. Moreover, Gao et al. [55] presented a modified road extraction network for multisource heterogeneous remote sensing data, which integrated local semantic features using the self-attention mechanism. In order to further promote the development of multimodal road extraction, Yang et al. [56] collected a large-scale real-world dataset for paired aerial image and trajectory data, covering over 33 000 km^2 in more than 80 cities. Then, they proposed a cross-check-based late-fusion strategy by seeking the optimum of each modality independently and then fusing the two results by further revisiting the over-killed trajectories with the guidance of aerial predictions. Liu et al. [18] fed trajectory graphs and RSIs into distinct networks for feature extraction. Then, they fused modular features from multiple layers to forecast the ultimate road. Despite the advancements made by

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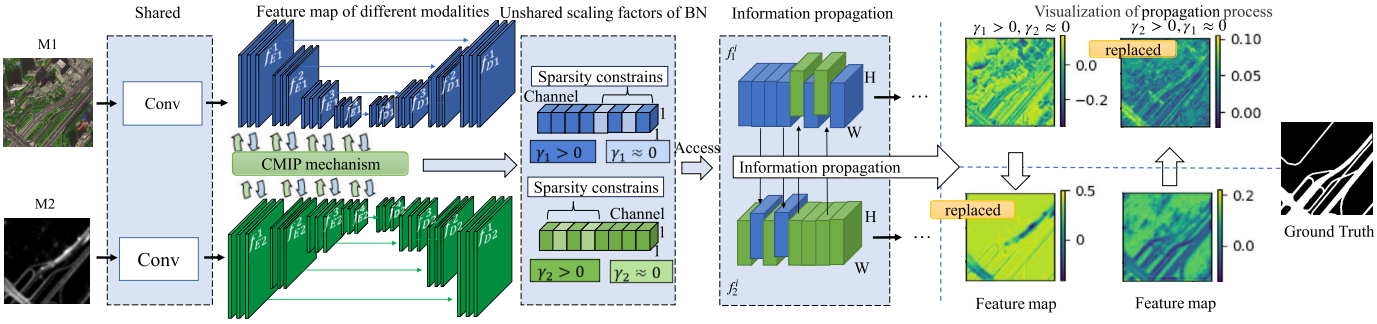


Fig. 2. Overall architecture of the proposed CMIPNet for road extraction. Our framework consists of 1) a dual-input, single-output encoder with shared convolutional parameters and 2) an information propagation mechanism for intra-modal feature interaction. The propagation process uses the scaling factor γ of independent BN to assess the importance of different modalities.

these approaches, the fusion manners do not fully utilize the complementarity of different modalities. Hence, Liu et al. [18] devised a dual-enhancement module to promote the interaction between trajectory graphs and RSIs. In addition, Bai et al. [57] designed a novel model to obtain road information accurately. They introduced an OR operation-based fusion strategy for combining image and trajectory data to extract road information, which can avoid the influence of trajectory noises on network training. Although these methods can moderately fuse information from different modalities, they fail to eliminate redundant information between modalities. Besides, most of their source code is not publicly available [53], [54], [55], [56], [57]. Hence, a more effective strategy is required to balance inter-modal information for enhanced road extraction.

C. Other Related Concepts

In earlier research endeavors, methods such as network pruning [24], [25] and representation learning [58] propose the idea of utilizing the scaling factors of the BN layer to assess the importance of CNN channels. Additionally, in a related study [24], a technique called "$\mathcal{L}1$ regularization" is applied to these scaling factors, which can introduce constraints to promote sparsity and facilitate the removal of specific redundant convolution filters. Drawing inspiration from this approach, we utilize the scaling factors as an adaptive tool to guide decisions on channel propagation and fusion. A similar approach can be observed in ShuffleNet [59], which suggests randomly shuffling channels among different groups to achieve efficient propagation. However, a key difference is that our information propagation process is determined by scaling factors of BN layers, in contrast to the stochastic channel shuffle employed in ShuffleNet. Besides, a recent study [60] showed that there existed redundancy between spectral information, which caused a high computational burden. Taking these into account, we propose information balanced constraints to reduce the redundant information between different modalities in a lightweight way.

III. Methodology

In this section, we introduce the framework of our CMIPNet and its two main components: the cross-modal feature propagation process and the information balanced constraints for feature selection.

A. Overall Framework

The overall framework of our CMIPNet is shown in Fig. 2. We first define the RSI and GPS trajectory (or LiDAR) modalities as $M1$ and $M2$, respectively. Then, we feed $M1$ and $M2$ into two encoder–decoder networks with the same architecture (D-LinkNet [14] is used in this article) to extract multimodal multiscale features. The encoding features of two modalities are denoted as $[f_{E1}^i, f_{E2}^i]$, where $E1$ and $E2$ represent two modality encoders, and $i \in \{1, 2, 3, 4\}$ means four encoding stages. Similarly, $[f_{D1}^i, f_{D2}^i]$ $(i \in \{4, 3, 2, 1\})$ represent the output decoding features. Specially, except for BN layers, the corresponding encoding and decoding layers of different modalities share the same parameters, which can not only reduce network complexity but also capture the common characteristics between the different modalities. On the other hand, the feature outputs of each modality are normalized using the independent BN layers, allowing the network to autonomously learn the modality-specific characteristics. This design facilitates the CMIP in subsequent stages. Besides, we embed the CMIP mechanism into each residual block of two encoders, whose illustration can be found in Fig. 2. The designed propagation mechanism first distinguishes the redundant information of each modality by assessing the importance of CNN channels based on the scaling factors of BN layers. Then, CMIP eliminates intra-modal noise channels by replacing them with more useful channels from another modality. The details of the CMIP mechanism will be analyzed in Section III-B. Moreover, a novel information balanced constraint on scaling factor is introduced to better identify and prune redundant channels, which will be presented in Section III-C.

In short, unlike previous methods based on direct or weighted fusion [17], [18], this work introduces a channel exchange-based fusion technique to achieve a simple yet effective interaction between two modal features. For each pair of $[f_{E1}^i, f_{E2}^i]$ from ith encoding stage, we employ information balanced constraints on the partial feature range of each modality to find the redundant channels and replace them by transferring useful channels from another modality using the propagation mechanism to enhance the robustness of feature representation. These enhanced $[f_{E1}^i, f_{E2}^i]$ are then fed into the next stages of two modality encoders, respectively, for further representation learning. Note that two modality decoders

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maintain the same structure as the decoder of D-LinkNet [14] without any modifications. The final output feature maps of two networks (i.e., f_{D1}^1 and f_{D2}^1) are employed together to predict a probability map denoted as $P \in R^{H\times W}$ using the following formulation:

$$P = \mathrm{Conv}\left(f_{D1}^1 \oplus f_{D2}^1, W_{3\times 3}\right) \tag{1}$$

where $\oplus$ denotes the feature concatenation operation, and $W_{3\times 3}$ represents the weight of the 3×3 convolutional layer. At each position (x, y) in P, a pixel is considered as a road pixel when its value is greater than the set threshold (0.5 in this article). Besides, in the propagation process (also shown in Fig. 2), γ_1 and γ_2 represent the scaling factors of two channels of $M1$ and $M2$, respectively.

In conclusion, the whole optimization goal of this model includes two aspects: 1) minimizing the error between the prediction maps and the ground-truth (GT) maps and 2) minimizing the constraints to maximize the network's ability to eliminate noise information while maintaining as much original modal information as possible, which is inspired by [24]. The overall optimization function can be formulated as

$$\min\left(\frac{1}{N}\sum_{i=1}^{N}\mathcal{L}\left(f\left(\boldsymbol{x}^{(i)}\right), \boldsymbol{y}^{(i)}\right) + \lambda\sum_{m=1}^{M}\sum_{l=1}^{L}R\left(\hat{\gamma}_{m,l}\right)\right) \tag{2}$$

in which $\min(1/N)\sum_{i=1}^{N}\mathcal{L}(f(\boldsymbol{x}^{(i)}), \boldsymbol{y}^{(i)})$ is used to optimize the above aspect 1), and $\min\sum_{m=1}^{M}\sum_{l=1}^{L}R(\hat{\gamma}_{m,l})$ can achieve the above aspect 2) (the detailed derivation of the proof can be found in [24]). In detail, $f(\boldsymbol{x}^{(i)})$ denotes the probability that the ith pixel of the predicted road map P represents the road, and $\boldsymbol{y}^{(i)}$ is the label of the corresponding pixel (1 for road, and 0 otherwise). $\mathcal{L}(x, \mathrm{y})$ is the loss function, and we use the combination of dice loss and BCE loss by default. Simultaneously, we apply additional BCE loss weights to the road edge regions. N represents training batch size. Moreover, M is the number of modalities, and L is the number of layers that require sparse constraints. $\gamma_{m,l}$ is the channel scaling factor of a certain modality m in the lth layer. $R(\cdot)$ denotes the designed information balanced constraint. We penalize the information balanced constraints of the certain portion of $\gamma_{m,l}$ (denoted $\hat{\gamma}_{m,l}$) and λ is a constant to control the penalty degree of it. These parameters are discussed in detail in Sections III-B and III-C.

Overall, using multimodal data for extracting road has already been established. However, revealing and solving the redundant information issue of multimodel feature maps is novel and under-explored. Our work can enhance road extraction performance by introducing additional modalities with only a slight increase in model size, surpassing the performance of using single-modal or other multimodal methods alone. Embedding the proposed CMIP mechanism into the most existing single- and multimodal models, with the only requirement being the use of independent BN layers, can bring significant performance improvement without bells and whistles. We believe that our CMIP mechanism can provide inspiration for other multimodal data processing tasks. Besides, multimodal road extraction from RSI and trajectory/LiDAR has been an emerging field in recent years, and compared to single-modal road extraction, there is still a lack of large-scale datasets. However, with the rapid increase in the number of vehicles on the roads, it is now feasible to gather a substantial amount of GPS trajectories from these vehicles to establish a comprehensive trajectory database [48], [49], [50]. With this crowdsourced mechanism, adequate trajectories can be easily collected nowadays.

B. Cross-Modal Information Propagation

Before delving into the information propagation process, let us revisit the BN layer [26], a common deep learning technique used to alleviate covariate shift and enhance generalization. In this context, let $\boldsymbol{x}_{m,l}$ represent the lth layer's feature map in the sub-network of modality m, and $\boldsymbol{x}_{m,l,c}$ represent the cth channel within it. The BN layer incorporates the following affine transformation:

$$\boldsymbol{x}'_{m,l,c} = \gamma_{m,l,c}\frac{\boldsymbol{x}_{m,l,c} - \mu_{m,l,c}}{\sqrt{\sigma_{m,l,c}^2 + \epsilon}} + \beta_{m,l,c}. \tag{3}$$

In this equation, $\mu_{m,l,c}$ and $\sigma_{m,l,c}$ represent the mean and standard deviation computed across all activations for the current mini-batch data over pixel locations. $\gamma_{m,l,c}$ and $\beta_{m,l,c}$ are trainable scaling factors and offset parameters, respectively. The small constant ϵ is introduced to prevent divisions by zero. Subsequently, the $(l+1)$th layer takes $\{\boldsymbol{x}'_{m,l,c}\}_c$ as input following a nonlinear transformation.

Specially, the scaling factor $\gamma_{m,l,c}$ measures the correlation between the input $\boldsymbol{x}_{m,l,c}$ and the output $\boldsymbol{x}'_{m,l,c}$ in the training process. If $\gamma_{m,l,c} \to 0$, the gradient of the loss with respect to $\boldsymbol{x}_{m,l,c}$ approaches 0. This indicates that $\boldsymbol{x}_{m,l,c}$ loses its influence on road predicting and becomes redundant in this task. This aligns with the concepts found in the network pruning [24], [25], where scaling factors of the BN layer can assess the importance of CNN channels. This idea inspires us to find the redundant channels with small BN scaling factors for a given modality and replace them with the channels from another modality located in the same position, thereby eliminating the redundant information and achieving the interaction of two modalities. Therefore, we derive

$$\boldsymbol{x}'_{m,l,c} = \begin{cases} \gamma_{m,l,c}\dfrac{\boldsymbol{x}_{m,l,c} - \mu_{m,l,c}}{\sqrt{\sigma_{m,l,c}^2 + \epsilon}} + \beta_{m,l,c}, & \text{if } \gamma_{m,l,c} > \theta \\ \gamma_{\tilde{m},l,c}\dfrac{\boldsymbol{x}_{\tilde{m},l,c} - \mu_{\tilde{m},l,c}}{\sqrt{\sigma_{\tilde{m},l,c}^2 + \epsilon}} + \beta_{\tilde{m},l,c}, & \text{else} \end{cases} \tag{4}$$

where the redundant channel of one modality (m) is replaced with the corresponding channel from another modality $(\tilde{m})$ when its corresponding scaling factor is under a threshold $\theta \approx 0^+$. We apply (4) to each f_{E1}^i and f_{E2}^i $(i \in \{1, 2, 3, 4\})$ before passing them through the nonlinear activation. Gradients are detached from the replaced channel and propagate backward through the newly introduced channels.

Now let us discuss why this information propagation can make the final optimization result better. Before explaining the effectiveness of information dissemination, it is crucial to establish that the performance of network can be maintained

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consistently through self-learning after information propagation, even if it has not learned any valuable cross-modal information. In detail, our optimization goal is to minimize (2), which implies that the training result pushes small $\hat{\gamma}_{m,l}$ to 0. Here, we denote the network functions with and without the implementation of the propagation mechanism as f' and f, respectively. For any function f, we can learn a specific function f' which produces identical outputs to f when $\hat{\gamma}_{m,l} = 0$. In the case of f, the architecture includes a BN layer followed by a convolutional layer. Given the input $\boldsymbol{x}_{m,l,c}$, we perform BN in the lth layer and get $\boldsymbol{x}'_{m,l,c}$. We denote the convolutional weights of the cth input channel of $\boldsymbol{x}'_{m,l,c}$ in the $(l+1)$th layer as $\boldsymbol{W}_{m,l+1,c}$ and the bias as $b_{m,l+1}$. Consequently, the convolution expression in the $(l+1)$th layer can be expressed as $(\boldsymbol{W}_{m,l+1,c} * \text{Relu}(\boldsymbol{x}'_{m,l,c}) + b_{m,l+1})$, in which $*$ denotes convolution operation. Given that $\gamma_{m,l,c} = 0$, this expression can be simplified to $(\boldsymbol{W}_{m,l+1,c} * \text{Relu}(\beta_{m,l,c}) + b_{m,l+1})$, resulting in a constant feature map. For channel characteristic $\boldsymbol{x}'_{m,l,c}$ corresponding to $\gamma_{m,l,c} \approx 0^+$ in f', we employ the similar notations and implement information propagation mechanism to obtain the term associated with $\boldsymbol{x}'_{\tilde{m},l,c}$ in the $(l+1)$th layer as $(\boldsymbol{W}'_{m,l+1,c} * \text{Relu}(\boldsymbol{x}'_{\tilde{m},l,c}) + b'_{m,l+1})$. By setting $\boldsymbol{W}'_{m,l+1,c} = 0$ and $b'_{m,l+1} = \boldsymbol{W}_{m,l+1,c} * \text{Relu}(\beta_{m,l,c}) + b_{m,l+1}$, we can always ensure $f' = f$ which means that even if the network does not learn any cross-modal information ($\boldsymbol{W}'_{m,l+1,c} = 0$), it can autonomously set $b'_{m,l+1}$ to ensure that $f' = f$. This indicates that after information propagation, f' can maintain the network's performance compared with f.

The above analysis confirms that as $\hat{\gamma}_{m,l}$ approaches 0, we can learn a mapping function f' that produces the same output results even without learning any useful cross-modal information. In other words, assuming that the network fails to acquire meaningful information after applying the information propagation mechanism, i.e., the replaced features are also redundant, its performance can be still maintained without degradation. Moreover, according to [59], the channel interchange across different groups can enable cross-group information flow for convolution layers and boost the network performance. This also applies to cross-modal information flow, which means that the cross-modal channel exchange can enhance the network's capacity for multimodal data representation. Therefore, we can conclude that by embedding information propagation, the network can acquire more valuable cross-modal information, which means that the loss can be further reduced, leading to $\min L_{f'} \leq \min L_f$ given the sufficiently expressive f' and f. In summary, we have proven that information propagation is helpful in improving the expression ability of the network and makes the final optimization result better.

C. Information Balanced Constraint

In order to make the network better filter the redundancy features of each modality, we add the sparse constraints on scaling factors. A commonly used sparse constraint is $\mathcal{L}1$ constraint which is utilized to identify and prune redundant channels in [24]. However, this type of constraint tends to push all scaling factors toward zero including those with relatively large values, because the optimal solution of $\min \|\gamma\|_1$ is $\mathbf{0}$. To obtain a better effect, we need to improve $\mathcal{L}1$ constraint by driving the smaller scaling factors toward zero while applying minimal constraints for the larger values. We name this constraint as information balanced constraint.

Let $\boldsymbol{\gamma}_{m,l} \in \mathbf{R}^n$ represents the lth layer's scaling factors of the mth modality sub-network and $\gamma_{m,l,c}$ denotes scaling factor of the cth channel, where $c \in [0, n]$ (n is the number of channels). Let $\mathbf{1}_n = (1, 1, \ldots, 1) \in \mathbf{R}^n$ and $\bar{\gamma}_{m,l}$ denote the mean value of $[\gamma_{m,l,1}, \ldots, \gamma_{m,l,n}]$

$$\bar{\gamma}_{m,l} = \frac{1}{n}\mathbf{1}_n^\top \boldsymbol{\gamma}_{m,l} = \frac{1}{n}\sum_{i=1}^{n} \gamma_{m,l,i}. \tag{5}$$

Then, we can define the information balanced constraints as

$$\begin{aligned} R(\boldsymbol{\gamma}_{m,l}) &= t\|\boldsymbol{\gamma}_{m,l}\|_1 - \|\boldsymbol{\gamma}_{m,l} - \bar{\gamma}_{m,l}\mathbf{1}_n\|_1 \\ &= \sum_{i=1}^{n} t|\gamma_{m,l,i}| - |\gamma_{m,l,i} - \bar{\gamma}_{m,l}|, \quad (t \in \mathbb{R}). \end{aligned} \tag{6}$$

In (6), we add a new term $-\|\boldsymbol{\gamma}_{m,l} - \bar{\gamma}_{m,l}\mathbf{1}_n\|_1$ to the original $\mathcal{L}1$ constrains. The effect of $-\|\boldsymbol{\gamma}_{m,l} - \bar{\gamma}_{m,l}\mathbf{1}_n\|_1$ is to separate $\gamma_{m,l,i} (1 \leq i \leq n)$ from their average as far as possible. In other words, minimizing these constraints can maximize the ability of penalizing smaller scale factors toward zero to uncover redundant channels while preventing larger scale factors from being penalized to zero. We also employ the parameter t to regulate the percentage of scaling factors that are constrained to be 0 within the constraint term.

Moreover, to prevent redundant information propagation between modalities, we divide the whole channels of each modality into two parts, and only add constraints on one-half channels of one modality and another half-channels of another modality. We denote the partial scale factors $\gamma_{m,l}$ under constraints as $\hat{\gamma}_{m,l}$. By utilizing this partial constraints strategy, the network can learn and uncover redundant information in the constrained channels, and then facilitate the flow of valuable road information from the unconstrained channels toward the constrained channels i.e., $\gamma_{m,l,c} \rightarrow \hat{\gamma}_{\tilde{m},l,c}$. This can not only preserve modal-specific information in the unconstrained part, but also avoid invalid propagation since the scaling factor of the propagation information in this part is out of the sparsity constraints. As a result, during channel information propagation, the transmission of information can be ensured to consist primarily of useful rather than redundant information.

To illustrate the benefit of this strategy, we visualize the channel scaling factors of a certain layer in Fig. 3. These three sub-figures represent the distribution of scaling factors of each modality under different kinds of constraints. The horizontal axis of Fig. 3 represents the different channels in the BN layer. The length of each bar reflects the proportion of scaling factors of the different modalities, with longer bars corresponding to more significant channels. Hence, it is evident that the information across channels of different modalities without constraints is relatively balanced. When we apply the constraints to all channels, all scaling factors can be suppressed. This implies that the information from two modalities may be redundant at the same channel position, leading to that the scale factors of two modalities tend to

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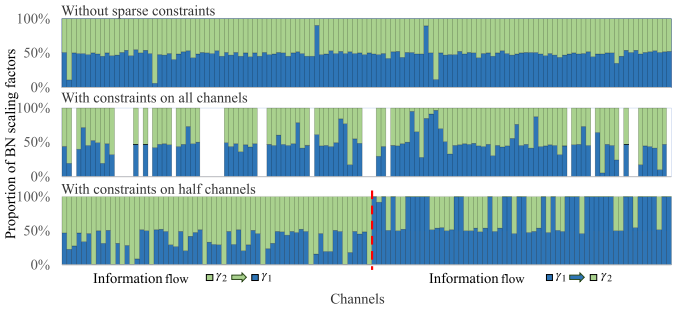


Fig. 3. Proportion of scaling factors γ for various channels of different modalities. They are calculated by $\gamma_{m,l,c}\ /(\gamma_{1,l,c}+\gamma_{2,l,c})$, in which m is 1 or 2 related to different modalities. (Top) Without sparsity constraints. (Middle) All channels of the lth layer are used to conduct information balanced constraints. We employ white space to denote a channel "c" when scaling factors ($\gamma_{1,l,c}$ and $\gamma_{2,l,c}$) are below the information propagation threshold. (Bottom) Information balanced constraints are applied to half of the channels of each modality.

approach zero simultaneously, as shown in the white blank part of Fig. 3 (middle). In this case, determining the direction of information propagation becomes challenging according to (4), and propagating information between redundant channels of two modalities is meaningless. When we implement constraints on the half-channels of each modality, the network can guarantee the transmission of only useful information between modalities. In detail, as illustrated in Fig. 3 (bottom), information propagates from channels with higher γ values to those with lower values. This means that channels in the left half transfer information from modality 1 to modality 2, while channels in the right half propagate information in the opposite direction.

We employ the above constraints and propose a more reliable strategy to automatically establish the propagation threshold. We show the distributions of scaling factors without/with constraint in Fig. 4. With information balanced constraint, it is clear that a majority of small scaling factors of a modality tend to be pushed toward ~0, while the original relatively large factors only have small fluctuations. Additionally, in contrast to $\mathcal{L}1$ constraints, the proposed balanced constraints strategy effectively defines propagation thresholds without necessitating manual threshold determination. Moreover, due to the effect of information balanced constraints, the distribution consistently exhibits at least two local maxima (peaks): one centered around 0 and others centered at larger values, as illustrated in the bottom of Fig. 4. Our approach focuses on replacing the channels with the peak closest to 0. Consequently, the threshold is precisely located at the tail of this nearest peak. To elaborate, when constructing the distribution histogram, we employ a bin width of 0.01. We scan the histogram bins from left to right and identify the first local minimum bin. The horizontal coordinate of this bin serves as the precise propagation threshold, as shown in Fig. 4.

Finally, let us analyze why information that is considered "redundant" and replaced must be worthless for model training. By utilizing the partial channel constraints strategy, the unconstrained channel scaling factors are necessarily greater than the constrained channel scaling factors. This ensures that information either propagates from unconstrained channels

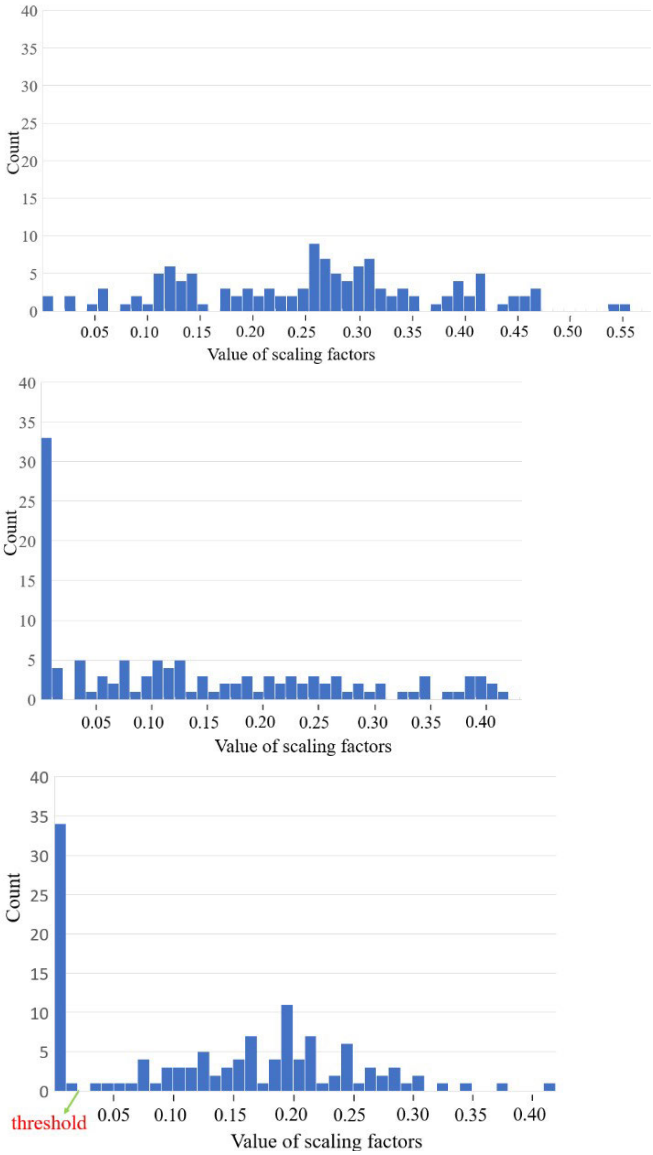


Fig. 4. Histogram distribution of channel scale factors under different constraints. Three sub-figures from top to bottom refer to the distributions of scaling factors without sparsity constraints, with $\mathcal{L}1$ constraints, and with our information balanced constraints, respectively.

to constrained channels, or does not propagate at all (when both channel scaling factors are high). Channels with very small scaling factors can be considered redundant since their influence on gradient propagation is negligible. Moreover, we visually display the evolution of channel scaling factors in a specific layer during training in Fig. 5. It is clear that the values of most high BN scaling factors are essentially stable, while the values of some BN scaling factors continue to decrease without subsequent increments (below the threshold or even close to zero). Consequently, we can confidently replace these redundant channels (i.e., channels with the BN scaling factors below the threshold) without concern that they re-provide the valuable information in subsequent training stages.

D. Comparison Between Our CMIPNet and CMMPNet

Overall, our CMIPNet and CMMPNet use the same network, i.e., D-LinkNet [14], as the backbone. Besides, the

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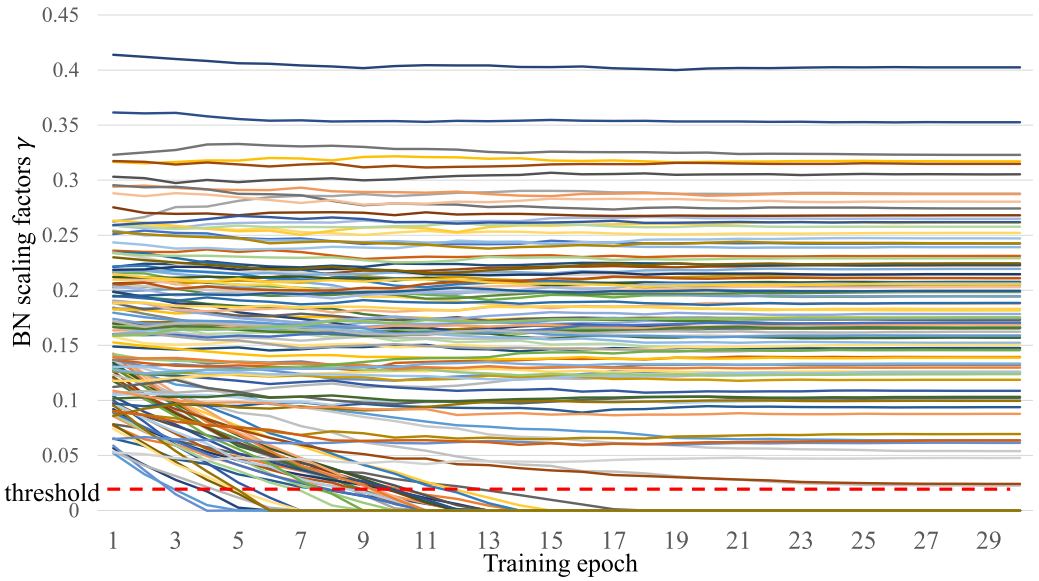


Fig. 5. Visualization of the change of BN scaling factors during the training process.

training strategies of them are also the same, which is provided in Section IV-A. On the other hand, the main differences between our CMIPNet and CMMPNet include the following.

1) CMMPNet uses two independent D-LinkNet networks to extract two modal features, respectively, while our CMIPNet uses the two parameter-shared D-LinkNet networks, which can effectively reduce the model complexity.
2) CMMPNet utilizes a dual-enhancement module (DEM) to fuse the features of different modalities mutually with a message-passing mechanism. However, this fusion module is essentially an attention-based gate mechanism, which ignores the serious negative impact of redundant information in different modalities. Our CMIPNet model reveals this phenomenon and designs a novel simple yet effective multimodal fusion strategy (i.e., CMIP mechanism) to deal with this issue. The different multimodel fusion strategies are the key differences between these two models.
3) The specific training hyperparameters of CMMPNet and CMIPNet are different, which are set experimentally. For example, CMMPNet utilizes a learning rate of 1e-4 and applies multiplicative decay when the training effect remains stable, while our CMIPNet is trained with a cosine decay strategy that initially uses an initial learning rate of 2e-4. The details can be found in Section IV-A.

IV. Experiments and Results

A. Experimental Setup

1) Datasets: The proposed method is evaluated on three publicly available multimodal road extraction datasets.

1) The BJRoad dataset, as described in [17], encompasses 1350 high-resolution RSIs covering approximately 100 km^2. It also includes approximately 50 million trajectory records from 28 000 vehicles. The images have a resolution of 1024 $\times$ 1024, with each pixel representing a real-world area of 0.5 $\times$ 0.5 m. Following previous work [18], we resize each image to the size of 512 $\times$ 512 to save the training memory occupation. The dataset is officially divided into three subsets: 70% for training, 10% for validation, and the remaining 20% for testing.
2) The TLCGIS dataset [21] comprises a total of 5860 pairs of aerial and LiDAR images. These images possess a resolution of 500 $\times$ 500, with each pixel representing a geographical length of 0.5 feet. The dataset is officially divided into training, testing, and validation sets, each containing 2640, 2400, and 240 samples, respectively.
3) The Porto dataset [19] encompasses an approximately 209-km^2 region in Porto, Portugal. This dataset comprises crowdsourced trajectories collected from 442 taxis between 2013 and 2014. Due to the absence of specific training/testing set details in [19], we split the complete RSI of the area into 6048 non-overlapping sub-images, each having a resolution of 512 $\times$ 512 pixels. Subsequently, these sub-images are randomly distributed into five equal sets. we allocate one-fifth of the dataset for testing purposes, with the remaining four-fifths for training.

2) Implementation Details: In this work, we implement CMIPNet on the representative PyTorch platform [61]. For the BJRoad dataset, we adopt the following data enhancement: flipping horizontally or vertically, rotating 90°, 180°, and 270°, and random cropping in the size range of [0.7, 0.9]. We also resize these images to a certain resolution, which is the same as previous work [18]. For the TLCGIS and Porto datasets, no data enhancement is carried out in this experiment for fair comparison. The proposed CMIPNet is trained for 30 epochs on BJRoad dataset and 50 epochs on TLCGIS and Porto datasets with a batch size of 8. Experimentally, the performance of CMIPNet tends to stabilize after 25 epochs on the validation set of BJRoad dataset and 45 epochs on the validation sets of TLCGIS and Porto datasets. In general, the training time of our CMIPNet is approximately 4 h. We set the learning

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rate to 0.0002 with the cosine annealing strategy and employ pre-trained weights from ResNet on ImageNet for all datasets. We use a Tesla V100 GPU with 16 GB memory to conduct our experiment. Finally, our work applies the Adam optimizer to optimize the loss between the generated road predicting map and the corresponding road GT map.

3) Evaluation Metrics: For a pixel(x, y), we determine it as a road region if the response prediction value of $P(x,y)$ surpasses 0.5. We use the Intersection over Union (IoU) and the graph-based metric of the average path length similarity (APLS) metrics to evaluate the segment performance of all models. Specially, the IoU metric quantifies the overlap between the road predicting map P_e and the GT P_g, which is computed by

$$\mathrm{IoU}\left(P_e, P_g\right) = \frac{|P_e \cap P_g|}{|P_e \cup P_g|} \tag{7}$$

where $|P_e \cap P_g|$ and $|P_e \cup P_g|$ denote the pixel number in the intersection set and union set of P_e and P_g, respectively.

Moreover, the APLS metric can measure the difference between the GT and the predicted graphs, which has been used to evaluate the road connectivity in the previous studies [62]. In detail, the APLS metric sums the differences in the optimal path lengths between nodes in the GT graph G and the predicted graph G'. The range of APLS metric is $[0, 1]$, and the larger the value, the better the performance. APLS can be calculated as follows:

$$\begin{aligned} S_{P\to T}\left(G, G'\right) &= 1 - \frac{1}{M}\sum \min\left(1, \frac{|L(a,b) - L\left(a',b'\right)|}{L(a,b)}\right) \\ S_{T\to P}\left(G', G\right) &= 1 - \frac{1}{M}\sum \min\left(1, \frac{|L(a,b) - L\left(a',b'\right)|}{L(a',b')}\right) \\ \mathrm{APLS} &= \frac{1}{N}\sum\left(\frac{1}{\frac{1}{S_{P\to T}} + \frac{1}{S_{T\to P}}}\right) \end{aligned} \tag{8}$$

where M and N represent the number of unique paths and images, respectively. The nodes a' and b' are the nodes in the predicted graph closest to the location of GT nodes a and b. $L(a, b)$ and $L(a', b')$ are the shortest path length of $a \to b$ in the GT graph and $a' \to b'$ in the predicted graph. $S_{P\to T}$ measures the sum of the difference of the shortest path for each node pair in the GT graph G and the predicted graph G'. $S_{T\to P}$ is similarly defined. Missing nodes of high centrality will be penalized much more heavily by the APLS metric than missing nodes of low centrality. Therefore, a high score can only be obtained if the road is properly connected.

Moreover, when using multimodal data, there are some important points to consider. First, for non-image modalities such as GPS data, it is necessary to convert them before inputting into the network. For example, after obtaining the raw GPS trajectories, we can transform them into the 2-D trajectory heatmaps [18]. In addition, to prevent the overfitting of the GPS trajectory modality, additional techniques should be employed, such as random subsampling of the input GPS data or ignoring random regions in the GPS data [17]. Furthermore, when incorporating additional modal information, the introduction of noise from other modalities is inevitable. To address this issue, our strategy utilizes an information propagation algorithm to allow the network to gradually reduce the noise from each modality during the learning process. By minimizing irrelevant noise within each modality, we can enhance the overall multimodal road extraction performance.

B. Performance Comparison

1) BJRoad Dataset [17]: As summarized in Table I, we compare our CMIPNet with 13 widely used approaches in terms of IoU and APLS metrics, including UNet [51], LinkNet [15], DeepLabv3 [63], Res-UNet [52], D-LinkNet [14], Sun et al. [17], DeepDualMapper [19], SA-Gate [64], CMMPNet [18], CMFormer [65], CMX [66], SGACNet [67], and CMNEXT [68]. Specially, the results of the top six methods are implemented by [17], and the results of DeepDualMapper [19] and CMMPNet [18] are implemented by [18]. SA-Gate [64], CMFormer [65], CMX [66], SGACNet [67], and CMNEXT [68] are five latest multimodal segmentation models with open-source codes. We re-train and re-evaluate them on the three datasets using the default codes released by the authors under the optimal settings. From Table I, observations reveal that the performance of multimodal models is generally better than that of single-modal models. Among the single-modal models, DeepLabv3 achieves the worst IoU of 50.81%, and D-LinkNet achieves the worst APLS of 49.91%. On the other hand, D-LinkNet achieves the best IoU of 57.96%, and LinkNet achieves the best APLS of 56.51%. These methods directly concatenate the channel-level data from RSIs and GPS trajectory heatmaps into the network. Their ability to capture and interact with the multimodal information is limited, thus failing to achieve an IoU above 60.00%. Among the multimodal models, DeepDualMapper [19] employs a gate fusion module to merge image and trajectory features, and attains competitive results with an IoU of 60.91%. CMMPNet [18] leverages the information from both RSIs and GPS trajectories to reinforce each other, and holds a significant advantage over previous methods. In detail, CMMPNet, implemented with D-LinkNet [14] as the backbone, achieves a highly competitive IoU of 62.85%. In general, our CMIPNet can achieve the best IoU value of 64.00%, which is significantly superior to other models. In terms of the APLS metric, our CMIPNet is also the best model. Hence, we can draw a conclusion that our simple channel feature propagation strategy can effectively reduce the influence of intra-/cross-modal noise and finally improve the performance to a new state of the art.

Additionally, we provide a visual comparison of our CMIPNet with the previous methods in Fig. 6 to illustrate the effective solution of our model to the above-mentioned issues. Overall, CMIPNet can achieve more accurate predictions by efficiently utilizing complementary multimodal information (the top three rows), and minimizing misjudgment of objects that share similar characteristics with roads (the last row). We can observe that the image-only model makes it difficult to accurately segment obstructed roads in RSIs. Besides, CMMPNet [18] is prone to misclassify objects that share

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TABLE I

COMPARISON BETWEEN OUR CMIPNET AND 13 STATE-OF-THE-ART METHODS IN TERMS OF IOU AND APLS ON THE TESTING SET OF THE BJROAD DATASET

Type	Method	Publication	IoU(%)	APLS(%)
Single-modal	UNet [51]	MICCAI'2015	54.88	53.81
	LinkNet [15]	VCIP'2017	57.89	56.51
	DeepLabv3+ [63]	ECCV'2018	50.81	54.33
	Res-UNet [52]	GRSL'2018	54.24	54.85
	D-LinkNet [14]	CVPR'2018	57.96	49.91
Multi-modal	Sun *et al.* [17]	CVPR'2019	59.18	55.83
	DeepDualMapper [19]	AAAI'2020	60.19	-
	SA-Gate [64]	ECCV'2020	62.14	57.78
	CMMPNet [18]	TNNLS'2022	62.85	54.28
	CMFormer [65]	PR'2023	63.45	53.28
	CMX [66]	TITS'2023	62.28	58.33
	SGACNet [67]	Sensors'2023	63.13	57.09
	CMNEXT [68]	CVPR'2023	63.22	56.01
	CMIPNet (ours)	–	**64.00**	**58.73**

* For single-modal models, we first concatenate the input images of different modalities along the channel dimension and feed the obtained inputs into the models.

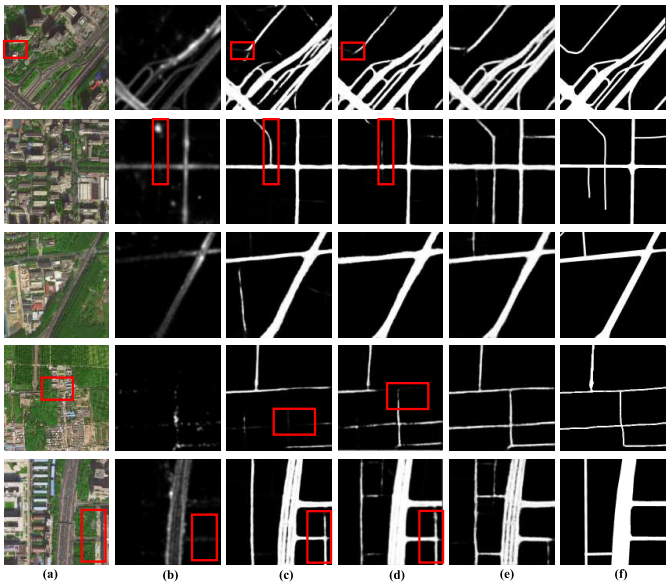


Fig. 6. Visual comparison of ours with the best competitors on BJRoad dataset. For the image-only methods, we adopt D-LinkNet to extract roads. We use the red box to highlight the regions with poor prediction results. (a) Image. (b) Trajectory. (c) Image-only. (d) CMMPNet. (e) Ours. (f) GT.

similar shapes with roads, while the prediction of our work is better in these aspects. However, from the last example of Fig. 6, our CMIPNet mistakenly extracted some roads. The main reason is as follows. We present detailed high-resolution images and corresponding trajectory maps of the road scene in the last example of Fig. 6 in Fig. 7. According to the blue box in the image, we can observe that these regions are maybe the narrow pathways that are obscured. Additionally, there exist some sparse corresponding GPS trajectories in these areas. Hence, our CMIPNet is confused and generates seemingly inaccurate results. On the contrary, the prediction based solely on the image does not include these two road segments because they do not use GPS trajectory information. Fortunately, there are very few cases where images/trajectories and truth maps do not completely match.

2) TLCGIS Dataset [21]: As displayed in Table II, we also compare CMIPNet with 12 existing state-of-the-art methods including SegNet [20], TriSeg [21], PreConv [21], RFNorm [21], ElePreConv [21], DeepDualMapper [19], SA-Gate [64], CMMPNet [18], CMFormer [65], CMX [66], SGACNet [67], and CMNEXT [68] on TLCGIS dataset in terms of IoU and APLS metrics. The details of these compared methods are described as follows. SegNet [20] is a fully convolutional autoencoder that takes remote sensing and LiDAR images as input. PreConv [21] processes LiDAR images through a depth convolution unit (DepthCNN) with two convolutional layers. The resulting LiDAR features are then concatenated with RSIs and fed into SegNet to extract roads. RFNorm [21] employs the random forest classifiers to estimate the road probability scores from aerial and LiDAR images. These score maps are concatenated and input into SegNet. In ElePreConv [21], LiDAR images undergo encoding with two convolutional layers, while images are extended with an additional zero-initialized channel. The element-wise addition of these images and LiDAR features is then fed into FuseNet. TriSeg [21] comprises three SegNets, the first two SegNets generate road probability maps from either aerial or LiDAR images, which are then concatenated and input into the third SegNet for final estimation. DeepDualMapper [41] is re-implemented by [18], which utilizes D-LinkNet as the backbone network. CMMPNet [18] integrates RSIs and LiDAR into two different backbone networks. Leveraging the proposed cross-modal information interaction module, it effectively combines the local and global information, thereby enhancing the fusion of these two data modalities. Overall, our CMIPNet also achieves a new state-of-the-art IoU (84.46%) and APLS (50.91%). We also provide a visualization of the results, as shown in Fig. 8.

3) Porto Dataset [19]: Furthermore, we conduct a performance comparison between our CMIPNet and eight competitive models, namely D-LinkNet [14], Sun et al. [17], SA-Gate [64], CMMPNet [18], CMFormer [65], CMX [66], SGACNet [67], and CMNEXT [68] on the Porto dataset. As presented in Table III, despite the already high benchmarks,

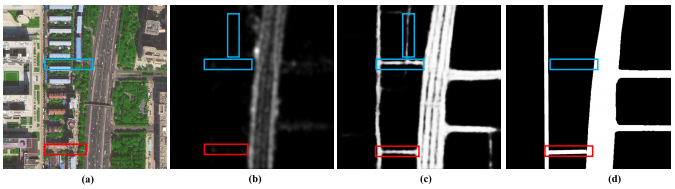


Fig. 7. Detailed high-resolution images of the last example in Fig. 6. (a) Image. (b) Trajectory. (c) Ours. (d) GT.

TABLE II

COMPARISON BETWEEN OUR CMIPNET AND 12 STATE-OF-THE-ART METHODS IN TERMS OF IOU AND APLS ON THE TESTING SET OF THE TLCGIS DATASET

Type	Method	Publication	IoU(%)	APLS(%)
Single-modal	SegNet [20]	TPAMI'2017	77.80	-
Multi-modal	TriSeg [21]	ACM SIGSPAI'2018	78.40	-
	PreConv [21]	ACM SIGSPAI'2018	71.30	-
	RFNorm [21]	ACM SIGSPAI'2018	75.20	-
	ElePreConv [21]	ACM SIGSPAI'2018	76.50	-
	DeepDualMapper [19]	CVPR'2020	81.10	-
	SA-Gate [64]	ECCV'2020	84.20	47.87
	CMMPNet [18]	TNNLS'2022	83.10	47.90
	CMFormer [65]	PR'2023	79.53	43.20
	CMX [66]	TITS'2023	84.14	45.89
	SGACNet [67]	Sensors'2023	84.11	47.25
	CMNEXT [68]	CVPR'2023	82.26	47.26
	CMIPNet (ours)	–	**84.46**	**50.91**

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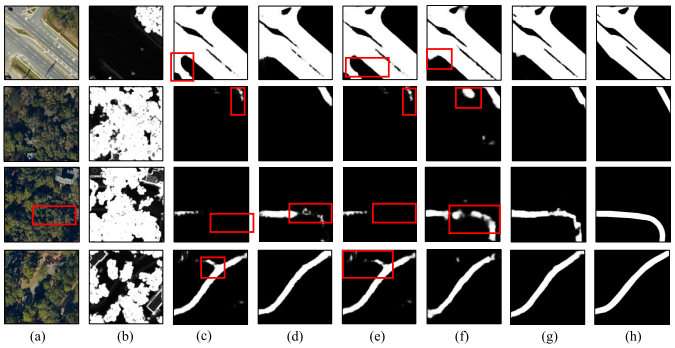


Fig. 8. Visual comparison of ours with the best competitors on TLCGIS dataset. For the image-only methods, we adopt D-LinkNet to extract roads. (a) Image. (b) LiDAR. (c) Image-only. (d) CMX. (e) SA-Gate. (f) SGACNet. (g) Ours. (h) GT.

TABLE III

COMPARISON BETWEEN OUR CMIPNET AND EIGHT METHODS IN TERMS OF IOU AND APLS ON THE TESTING SET OF THE PORTO DATASET

Type	Method	Publication	IoU(%)	APLS(%)
Single-modal	D-LinkNet [14]	CVPR'2018	70.20	70.64
Multi-modal	Sun *et al.* [17]	CVPR'2019	71.79	72.60
	SA-Gate [64]	ECCV'2020	72.21	72.59
	CMMPNet [18]	TNNLS'2022	72.29	72.52
	CMFormer [65]	PR'2023	70.08	70.13
	CMX [66]	TITS'2023	72.85	73.65
	SGACNet [67]	Sensors'2023	72.50	73.07
	CMNEXT [68]	CVPR'2023	73.12	**74.45**
	CMIPNet (ours)	–	**73.50**	73.71

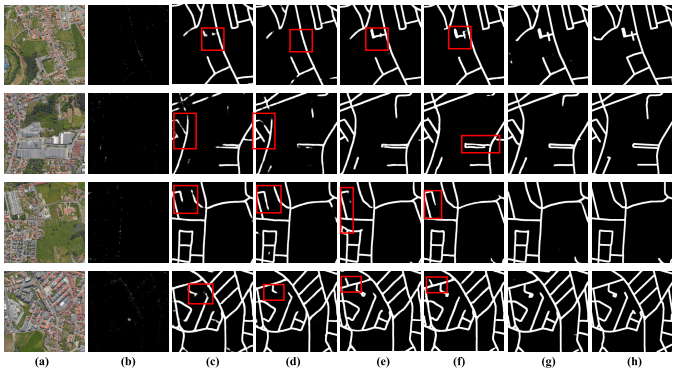


Fig. 9. Visual comparison of ours with the best competitors on Porto dataset. For the image-only methods, we adopt D-LinkNet to extract roads. (a) Image. (b) Trajectory. (c) Image-only. (d) CMMPNet. (e) CMX. (f) CMNEXT. (g) Ours. (h) GT.

CMIPNet significantly improves the IOU and achieves the top performance ranking on the Porto dataset. In terms of APLS, the performance of our CMIPNet is slightly lower than that of CMNEXT [68]. In conclusion, these comparisons strongly affirm the effectiveness of our proposed CMIPNet for road extraction based on image and trajectory data. We also provide visualization of the results in Fig. 9. It is clear that the situations of this dataset are more complex than those of the other two datasets. The roads in Porto dataset are usually thin, dense, and heavily interrupted by buildings. From Fig. 9, we can see that our model can still accurately extract road areas with finer details.

4) Practical Application Within Large Area: Due to the extensive spatial requirements of the input road data, most works divide the input images into segments to extract roads.

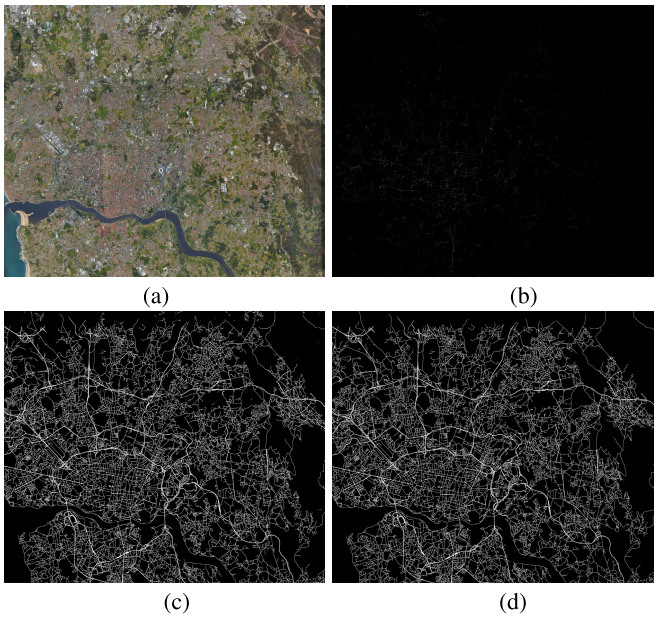


Fig. 10. Visualization of the large RSI, GPS trajectory, predicted map, and GT map. (a) RSI. (b) GPS trajectory. (c) Ours. (d) GT.

To verify the feasibility of the proposed method in the practical application within a large area, we generate the final merged road map as shown in Fig. 10. It can be seen that our CMIPNet still performs well visually. Moreover, the overall IoU value of the large road map extracted by our CMIPNet is 75.56%.

C. Effect of Multimodal Road Extraction

We initially investigate the reliability of utilizing multimodal data for road extraction in traffic scenarios. As demonstrated in Table IV, when feeding the trajectory heatmaps into the single-modal network D-LinkNet, we can obtain the subpar performance (IoU 52.38%) on the BJRoad dataset. Besides, using only RSIs as input obtains an IoU of 59.79%, which indicates the greater significance of image data compared to trajectory data. When utilizing both images and trajectory heatmaps as input, our CMIPNet achieves better performance (IoU 64.00% versus 62.8%) with lower parameters consumption (119.29 versus 328.74 M) compared to CMMPNet [18] that uses the same backbone. Thanks to the strategy of shared convolution, our network can surpass the previous dual-flow road extraction network by introducing only a little extra BN layer parameters. Hence, the number of parameters of our CMIPNet can be comparable to that of the single-modal model.

We also visualize the scaling factors in the output layers of the $[f_{E1}^i, f_{E2}^i]$ (i = 1,2,3,4) in Section III-A, which is shown in Fig. 11. We divide channels into two sub-parts and apply information balanced constraints on one sub-part for each modality. We can observe that as each stage of the information propagation mechanism is executed, certain modalities' scale factors tend to dominate in particular sections (in the left half of the channels, the green modality predominates, while in the right half, the blue modality takes precedence). This indicates that the network continuously learns redundant information from various modalities and achieves targeted information

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TABLE IV

PERFORMANCE OF DIFFERENT INPUTS AND DIFFERENT LEARNING METHODS ON THE TESTING SETS OF THREE DATASETS

Input	Method	Parameters	BJRoad		TLCGIS		Porto	
			IoU(%)	APLS(%)	IoU(%)	APLS(%)	IoU(%)	APLS(%)
Trajectory/LiDAR	D-LinkNet	119.04	52.38	45.31	69.12	36.87	40.86	37.77
Image	D-LinkNet	119.04	59.79	51.01	80.96	46.65	70.20	70.64
Image+Trajectory/LiDAR	D-LinkNet+CMMPNet	328.74	62.80	54.28	83.10	47.90	72.29	72.52
Image+Trajectory/LiDAR	D-LinkNet+CMIPNet	119.29	**64.00**	**58.73**	**84.46**	**50.91**	**73.50**	**73.71**

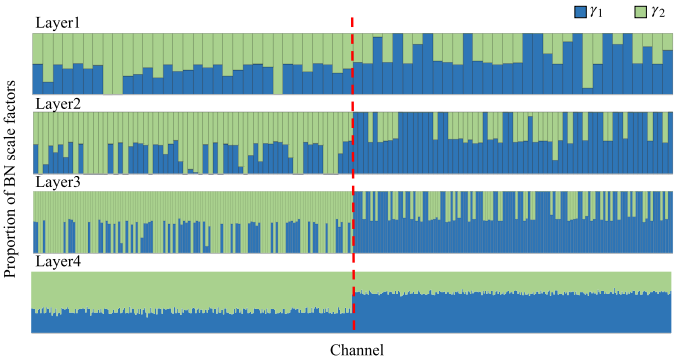


Fig. 11. Proportion of scaling factors in our network with backbone D-LinkNet. On the left side of the diagram, channels for modality γ_1 are constrained, while on the right side, channels for modality γ_2 are constrained. This directional constraint facilitates information propagation.

TABLE V

EFFECTIVENESS OF THE MAIN COMPONENTS IN OUR CMIPNET ON BJROAD DATASET

Convs	CMIP	Constraint	IoU (%)	APLS(%)
Unshared	✗	✗	60.78	49.92
Shared	✗	✗	62.16	54.30
Shared	✓	✗	62.74	56.51
Shared	✓	All-channel	63.50	58.48
Shared	✓	Half-channel	**64.00**	**58.73**

TABLE VI

REPLACING THE PROPOSED CMIP MECHANISM WITH OTHER FUSION STRATEGIES. THE EXPERIMENT IS CONDUCTED ON BJROAD DATASET

Fusion Strategy of	**Publication**	**IoU(%)**	**APLS(%)**
Default (ours)	–	**64.00**	**58.73**
SA-Gate [64]	ECCV'2020	62.49	58.10
CMMPNet [18]	TNNLS'2022	61.43	52.45
CMX [66]	TITS'2023	62.53	57.82

propagation (from unconstrained modalities to constrained redundant modalities). In the final layer (layer 4) of feature extraction, the contributions from each modality are relatively balanced. This also indicates that the network has roughly completed the filtering of redundant information within each modality.

D. Ablation Studies

We conduct various ablation experiments to demonstrate the effectiveness of the main components of our CMIPNet, and the results are shown in Table V. We have the following observations.

1) As shown in the first and second rows of Table V, compared with the model with unshared convolution structure (60.78%), the model with the shared convolution parameters (62.16%) greatly improves the performance.

2) The second to fourth lines of Table V show that 1) the propagation mechanism can promote the performance from 62.16% to 62.74%, 2) the IoU metric is improved to 63.50% after the information balanced constraint being introduced, and 3) after constraining only the half of the channels for each modality, the IoU is improved to 64.00% (the fifth line in Table V). The results indicate the effectiveness of the designed propagation mechanism and information balanced constraint. Moreover, to display the performance evolution of various ablation experimental models more intuitively, we further provide the visual comparison of the inference results of different models in Fig. 12. Note that the third to seventh column models of Fig. 12 correspond to the first to fifth row models of Table V. It can be seen that the visual performance of the ablation experimental models progressively improves with the gradual addition of the new components, indicating the effectiveness of all proposed components of CMIPNet.

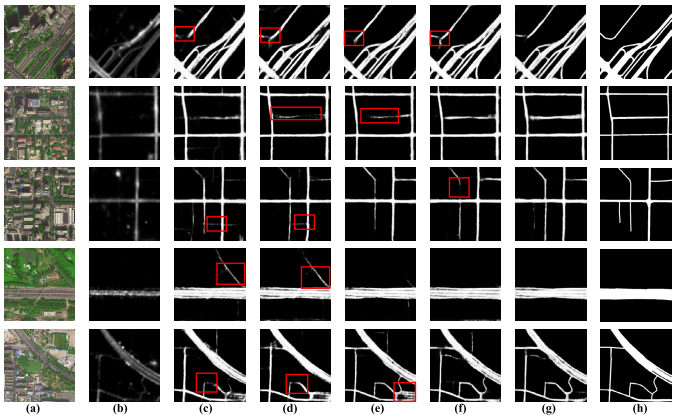


Fig. 12. Visual comparison of various ablation experimental models. (a) Image. (b) Trajectory. (c) Unshared. (d) Shared. (e) With CMIP. (f) All-channel. (g) Half-channel. (h) GT.

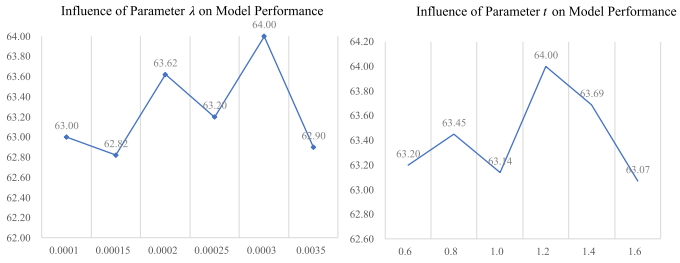


Fig. 13. Ablation studies for the parameters of CMIPNet, including λ and t in (2). λ is the weight of sparsity constraints, and t is a balancing factor for controlling the strength of $\mathcal{L}1$ regularization in the network.

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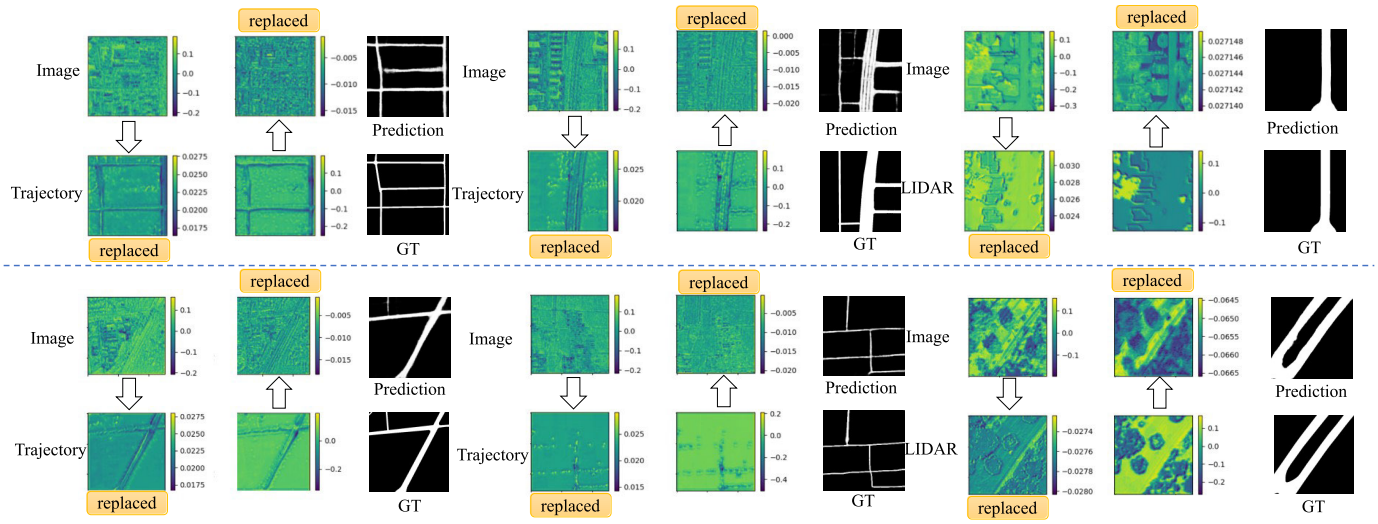


Fig. 14. Visualization of the averaged feature maps in different ranges γ for different modalities. The direction of the arrows is from one modality with $\gamma > 0$ to another modality with $\gamma \approx 0$.

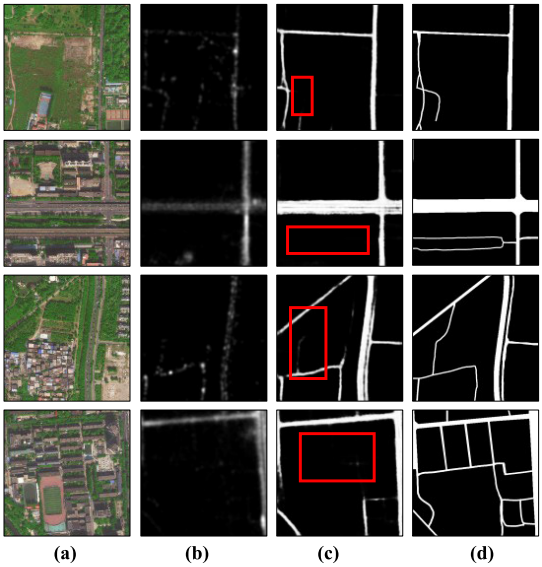


Fig. 15. Examples of the failure case visualization of our CMIPNet. (a) Image. (b) Trajectory. (c) Ours. (d) GT.

3) To explicitly demonstrate the effectiveness of our CMIP fusion mechanism, we replace the proposed CMIP with three fusion strategies of SA-Gate [64], CMMPNet [18], and CMX [66], respectively. The detailed comparison results on BJRoad dataset are shown in Table VI. It is clear that under the same network architecture and training strategy, the default CMIP fusion strategy achieves the best performance.

4) To explain how the default parameters of CMIPNet are set, we evaluate the performance with varying hyper-parameters, whose results are shown in Fig. 13. We first study the influence of the parameter λ, which is set from 1×10^{-4} to 3.5×10^{-4} (t is fixed to 1.2). It is evident that the model's performance exhibits an ascent followed by a descent as the parameter λ increases. This phenomenon is readily comprehensible: with the augmentation of λ, the model identifies and replaces more modality-specific redundant information. However, as λ continues to increase, the strength of the information balanced constraints proportionally intensifies. Consequently, the performance improvement attributable to information propagation becomes increasingly marginal in comparison to the regularization effect, leading to a gradual deterioration in the model's performance. For the experiments of parameter t, we set λ to 3×10^{-4} by default and vary t from 0.6 to 1.6. It can be observed that an excessively large value of t may cause the model to discard some valuable information, while a smaller value of t may lead to the challenge of replacing redundant information. In this work, we set t to 1.2.

E. Process of Information Propagation

In order to facilitate a deeper comprehension of how our model leverages CMIP to mitigate inter-modal redundancy and enhance road characteristics, we visualize the feature propagation process of certain layers, as depicted in Fig. 14. We can find that RSIs with non-zero scaling factors mainly encode texture information, while trajectory prioritizes road contours and LiDAR represents other objects that are not related to the road. In this regard, information propagation can enhance the fusion of the mutually complementary characteristics exhibited by two modalities. By performing the propagation process, the network has a tendency to selectively replace features with low activation values and minimal variation between locations, as well as those areas irrelevant to road contours. This phenomenon could be interpreted as a form of noise filtration across different modalities.

F. Failure Case Analysis

Although our CMIPNet achieves a new state of the art, it may still fail in some cases which are reported in Fig. 15. It is obvious that the majority of erroneous road prediction maps can be attributed to insufficient multimodal information, particularly in the first three rows. In the first rows, the model encounters difficulties in road identification due to tree obstructions and limited trajectory information, making

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it challenging to detect roads in these areas. In the second row, the red box highlights a railway with two roads below it. Moreover, the trajectory modality also cannot provide clear trajectory information, causing our CMIPNet to misjudge a road as a railway. Regarding the third row in Fig. 15, our CMIPNet struggles to identify the dirt road in the RSI. This issue is partly due to the fact that some images in the datasets tend not to label the dirt roads, resulting in inconsistent labeling that affects the performance of the model. In the last row, the model faces the challenges in predicting roads within complex urban neighborhoods. Furthermore, if other modalities can be introduced to provide sufficient details about narrow urban roads, the model can address this issue.

V. Conclusion

In this study, we address the challenging task of multimodal road extraction, especially how to leverage different modalities and reduce the negative impact of noise to learn road-related valuable information. To achieve this goal, we introduce the concept of network pruning and propose a novel cross-modal information propagation network, namely CMIPNet. CMIPNet employs the scaling factor of the BN layer as a metric to measure and replace the corresponding channels. Besides, we improve the $\mathcal{L}1$ sparse constraints and design an information balanced constraint to drive the smaller scaling factors toward zero while exerting minimal constraints on the larger values. The information propagation process can adapt itself dynamically through the control of scaling factors without adding additional parameters. Results show that our CMIPNet outperforms several state-of-the-art methods on three public multimodal datasets. We believe that such a paradigm can be widely applied in the future. A legacy issue in multimodal road extraction is that when the modal information is incomplete, multimodal road extraction still cannot solve the problems of road occlusion or misjudgment. For example, if there is no trajectory information or LiDAR signal for the blocked road, the feature propagation between modalities still cannot solve this situation.

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